

Stage 4 Rec-Del Pairing: Expected Output for Transco

Transcontinental Gas Pipe Line Company, LLC | TSP DUNS 007933021

Generated 2026-09-28 by running the stage 4 SQL in `src/transformations/stage_4/rec_del_pairing/pairing_base.py` against both Transco fixture files inside a rolled-back transaction on the Neon database, after the S8/S9 standardization. Nothing shown here was committed; the database tables are unchanged.

How to read this document

One row per receipt or delivery location. A contract with one receipt and one delivery is two rows. Storage locations (WR withdrawal, SB storage area, IJ injection) take no part and do not appear.

Receipt / delivery classification. M2 = Receipt Location and MQ = Delivery Location come from the feed itself. S9 = Receipt and S8 = Delivery is the Transco segment-end rule agreed with Aviral, applied by stage 3 standardization only to rows whose TSP DUNS is 007933021.

Formation is the contract's R/D letters in location (Index) order. **Pattern** is the row of the dashboard's Rec-Del Pairings table whose regex admitted that formation.

Set id restarts at 1 for every contract and is shared by the locations that belong together. Under an alternating pattern (regex of the form `^(R-D)+$`) each repetition is a set, so R-D-R-D reads 1,1,2,2. Under `R^n-D^m` the whole block is one set, so R-R-R-D-D-D reads 1,1,1,1,1,1.

Status is PAIRED when a pattern admitted the formation, otherwise FAIL (the rows still appear, with no set id).

Validating against the contract. The last column of each detail table says whether the location's name is listed in the contract's own RecLocs (for R rows) or DelLocs (for D rows) in the source JSON. Every named location says Yes. Six rows in Transco-F2 and Transco-F3 have no name at all in the JSON (location code 0, zone 4A only, zero quantity); they are marked *unnamed zone-level end* and are called out in Appendix B.

Fixture file	Contracts	Location rows in JSON	R/D rows in stage 4	PAIRED	FAIL
firm_transco_nationalgrid_10.json National Grid slice (served by /api/firms, used in the demo)	10	44	20	20	0
firm_transco_10.json Mixed slice (not wired to an endpoint)	10	67	46	46	0

firm_transco_nationalgrid_10.json

National Grid slice (served by /api/firms, used in the demo)

Summary: one line per contract

Contract	Contract holder	Contract Qty	Locations in JSON	Formation	Pattern	Status	Set ids	R/D rows
Transco-F24	BOSTON GAS COMPANY D/B/A NATIONAL GRID	6,911	2 locations; R/D codes: S9-S8	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F246	KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID	2,100	2 locations; R/D codes: S9-S8	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F247	KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F248	KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F249	KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F253	KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F460	THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F461	THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F462	THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F463	THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
TOTAL								20

Detail: every stage 4 row

Rows are grouped by contract; alternating shading separates contracts. Within a contract, rows sharing a set id are one pairing.

Contract	Set	Pos	R/D	Idx	Raw code	Loc code	Location name	Zone	Qty (Dth)	In contract RecLocs / DelLocs?
Transco-F24	1	1	RECEIPT	1	S9	1006325	WHARTON-NAT FUEL	ZONE 6		Yes
Transco-F24	1	2	DELIVERY	2	S8	1006561	ALGONQUIN-CTR M4402	ZONE 6	6,911	Yes
Transco-F246	1	1	RECEIPT	1	S9	1006161	LEIDY-DOMINION	ZONE 6		Yes

Contract	Set	Pos	R/D	Idx	Raw code	Loc code	Location name	Zone	Qty (Dth)	In contract RecLocs / DelLocs?
Transco-F246	1	2	DELIVERY	2	S8	1006382	LONG ISLAND LIGHTING	ZONE 6	2,100	Yes
Transco-F247	1	1	RECEIPT	4	M2	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F247	1	2	DELIVERY	5	MQ	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F248	1	1	RECEIPT	4	M2	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F248	1	2	DELIVERY	5	MQ	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F249	1	1	RECEIPT	4	M2	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F249	1	2	DELIVERY	5	MQ	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F253	1	1	RECEIPT	4	M2	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F253	1	2	DELIVERY	5	MQ	1006382	LONG ISLAND LIGHTING	ZONE 6		Yes
Transco-F460	1	1	RECEIPT	4	M2	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F460	1	2	DELIVERY	5	MQ	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F461	1	1	RECEIPT	4	M2	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F461	1	2	DELIVERY	5	MQ	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F462	1	1	RECEIPT	4	M2	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F462	1	2	DELIVERY	5	MQ	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F463	1	1	RECEIPT	4	M2	1006558	BROOKLYN UNION	ZONE 6		Yes
Transco-F463	1	2	DELIVERY	5	MQ	1006558	BROOKLYN UNION	ZONE 6		Yes

firm_transco_10.json

Mixed slice (not wired to an endpoint)

Summary: one line per contract

Contract	Contract holder	Contract Qty	Locations in JSON	Formation	Pattern	Status	Set ids	R/D rows
Transco-F1	AGY HOLDING, CORP.	341	8 locations; R/D codes: S9-S8-S9-S8-S9-S8-S9-S8	R-D-R-D-R-D-R-D	Alternating R-D	PAIRED	1, 1, 2, 2, 3, 3, 4, 4	8
Transco-F2	ANGOLA LNG SUPPLY SERVICES LLC	467,390	6 locations; R/D codes: S9-S8-S9-S8-S9-S8	R-D-R-D-R-D	Alternating R-D	PAIRED	1, 1, 2, 2, 3, 3	6
Transco-F3	ANGOLA LNG SUPPLY SERVICES LLC	467,390	6 locations; R/D codes: S9-S8-S9-S8-S9-S8	R-D-R-D-R-D	Alternating R-D	PAIRED	1, 1, 2, 2, 3, 3	6
Transco-F8	ATLANTA GAS LIGHT COMPANY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F9	ATLANTA GAS LIGHT COMPANY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F10	ATLANTA GAS LIGHT COMPANY		5 locations; R/D codes: M2-MQ; storage: IJ, SB, WR	R-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1	2
Transco-F176	DELMARVA POWER & LIGHT COMPANY		7 locations; R/D codes: M2-M2-MQ-MQ; storage: IJ, SB, WR	R-R-D-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1, 1, 1	4
Transco-F182	DOMINION ENERGY SOUTH CAROLINA INC		9 locations; R/D codes: M2-M2-M2-MQ-MQ-MQ; storage: IJ, SB, WR	R-R-R-D-D-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1, 1, 1, 1, 1	6
Transco-F337	PATRIOTS ENERGY GROUP		9 locations; R/D codes: M2-M2-M2-MQ-MQ-MQ; storage: IJ, SB, WR	R-R-R-D-D-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1, 1, 1, 1, 1	6
Transco-F513	VIRGINIA NATURAL GAS INC		7 locations; R/D codes: M2-M2-MQ-MQ; storage: IJ, SB, WR	R-R-D-D	R^n-D^m (n>=1,m>=1)	PAIRED	1, 1, 1, 1	4
TOTAL								46

Detail: every stage 4 row

Rows are grouped by contract; alternating shading separates contracts. Within a contract, rows sharing a set id are one pairing.

Contract	Set	Pos	R/D	Idx	Raw code	Loc code	Location name	Zone	Qty (Dth)	In contract RecLocs / DelLocs?
Transco-F1	1	1	RECEIPT	1	S9	1000007	STA 30 POOL ZN1	ZONE 1		Yes
Transco-F1	1	2	DELIVERY	2	S8	9003140	CAROLINA GAS	ZONE 5	58	Yes
Transco-F1	2	3	RECEIPT	3	S9	1000026	STA 45 POOL ZN2	ZONE 2		Yes
Transco-F1	2	4	DELIVERY	4	S8	9003140	CAROLINA GAS	ZONE 5	85	Yes
Transco-F1	3	5	RECEIPT	5	S9	1000040	STA 50 POOL ZN3	ZONE 3		Yes
Transco-F1	3	6	DELIVERY	6	S8	9003140	CAROLINA GAS	ZONE 5	65	Yes
Transco-F1	4	7	RECEIPT	7	S9	1000762	STA 62 POOL ZN3	ZONE 3		Yes
Transco-F1	4	8	DELIVERY	8	S8	9003140	CAROLINA GAS	ZONE 5	133	Yes

Contract	Set	Pos	R/D	Idx	Raw code	Loc code	Location name	Zone	Qty (Dth)	In contract RecLocs / DelLocs?
Transco-F2	1	1	RECEIPT	1	S9	9006145	MOSS POINT M3593	ZONE 4A		Yes
Transco-F2	1	2	DELIVERY	2	S8	9005542	STA 85 POOL ZN4A	ZONE 4A	467,390	Yes
Transco-F2	2	3	RECEIPT	3	S9	0	(none in JSON)	ZONE 4A		unnamed zone-level end
Transco-F2	2	4	DELIVERY	4	S8	0	(none in JSON)	ZONE 4A	0	unnamed zone-level end
Transco-F2	3	5	RECEIPT	5	S9	0	(none in JSON)	ZONE 4A		unnamed zone-level end
Transco-F2	3	6	DELIVERY	6	S8	1000105	STA 85 POOL ZN4	ZONE 4	0	Yes
Transco-F3	1	1	RECEIPT	1	S9	9006145	MOSS POINT M3593	ZONE 4A		Yes
Transco-F3	1	2	DELIVERY	2	S8	9005542	STA 85 POOL ZN4A	ZONE 4A	467,390	Yes
Transco-F3	2	3	RECEIPT	3	S9	0	(none in JSON)	ZONE 4A		unnamed zone-level end
Transco-F3	2	4	DELIVERY	4	S8	0	(none in JSON)	ZONE 4A	0	unnamed zone-level end
Transco-F3	3	5	RECEIPT	5	S9	0	(none in JSON)	ZONE 4A		unnamed zone-level end
Transco-F3	3	6	DELIVERY	6	S8	1000105	STA 85 POOL ZN4	ZONE 4	0	Yes
Transco-F8	1	1	RECEIPT	4	M2	1003410	ATLANTA GAS LIGHT	ZONE 4		Yes
Transco-F8	1	2	DELIVERY	5	MQ	1003410	ATLANTA GAS LIGHT	ZONE 4		Yes
Transco-F9	1	1	RECEIPT	4	M2	1003410	ATLANTA GAS LIGHT	ZONE 4		Yes
Transco-F9	1	2	DELIVERY	5	MQ	1003410	ATLANTA GAS LIGHT	ZONE 4		Yes
Transco-F10	1	1	RECEIPT	4	M2	1003410	ATLANTA GAS LIGHT	ZONE 4		Yes
Transco-F10	1	2	DELIVERY	5	MQ	1003410	ATLANTA GAS LIGHT	ZONE 4		Yes
Transco-F176	1	1	RECEIPT	4	M2	1006570	EASTERN SHORE	ZONE 6		Yes
Transco-F176	1	2	RECEIPT	5	M2	1006587	DELMARVA	ZONE 6		Yes
Transco-F176	1	3	DELIVERY	6	MQ	1006570	EASTERN SHORE	ZONE 6		Yes
Transco-F176	1	4	DELIVERY	7	MQ	1006587	DELMARVA	ZONE 6		Yes
Transco-F182	1	1	RECEIPT	4	M2	1003399	ABBEVILLE-SCEG M7166	ZONE 5		Yes
Transco-F182	1	2	RECEIPT	5	M2	1006586	GREENWOOD	ZONE 5		Yes
Transco-F182	1	3	RECEIPT	6	M2	9003140	CAROLINA GAS	ZONE 5		Yes
Transco-F182	1	4	DELIVERY	7	MQ	1003399	ABBEVILLE-SCEG M7166	ZONE 5		Yes
Transco-F182	1	5	DELIVERY	8	MQ	1006586	GREENWOOD	ZONE 5		Yes
Transco-F182	1	6	DELIVERY	9	MQ	9003140	CAROLINA GAS	ZONE 5		Yes
Transco-F337	1	1	RECEIPT	4	M2	1003399	ABBEVILLE-SCEG M7166	ZONE 5		Yes
Transco-F337	1	2	RECEIPT	5	M2	1006586	GREENWOOD	ZONE 5		Yes
Transco-F337	1	3	RECEIPT	6	M2	9003140	CAROLINA GAS	ZONE 5		Yes
Transco-F337	1	4	DELIVERY	7	MQ	1003399	ABBEVILLE-SCEG M7166	ZONE 5		Yes

Contract	Set	Pos	R/D	Idx	Raw code	Loc code	Location name	Zone	Qty (Dth)	In contract RecLocs / DelLocs?
Transco-F337	1	5	DELIVERY	8	MQ	1006586	GREENWOOD	ZONE 5		Yes
Transco-F337	1	6	DELIVERY	9	MQ	9003140	CAROLINA GAS	ZONE 5		Yes
Transco-F513	1	1	RECEIPT	4	M2	1000941	EMPORIA WALDROP	ZONE 5		Yes
Transco-F513	1	2	RECEIPT	5	M2	1006173	LYNCHBURG	ZONE 5		Yes
Transco-F513	1	3	DELIVERY	6	MQ	1000941	EMPORIA WALDROP	ZONE 5		Yes
Transco-F513	1	4	DELIVERY	7	MQ	1006173	LYNCHBURG	ZONE 5		Yes

Appendix A: the pattern gate (public.rec_del_pairings)

The dashboard's Rec-Del Pairings table as it stood when this was generated. Stage 4 strips the dashes from both the formation and the regex before matching, so $^(R-D)+\$$ is tested as $^(RD)+\$$ against RDRD. Without that, the alternating regexes could only match a single R-D and every alternating contract came back FAIL. The unit column is what stage 4 derives: locations per set, or 'whole formation' when the regex has no repeating group.

Order	Pipeline	DUNS	Pattern	Regex	Set unit derived by stage 4	Admits (formations seen in these files)
1	default	0	R^n-D^m ($n \geq 1, m \geq 1$)	$^R(?:-R)^*-D(?:-D)^*\$$	whole formation = 1 set	R-D, R-R-D-D, R-R-R-D-D-D
2	default	0	D^n-R^m ($n \geq 1, m \geq 1$)	$^D(?:-D)^*-R(?:-R)^*\$$	whole formation = 1 set	-
3	default	0	Alternating R-D	$^(R-D)+\$$	2 locations per set	R-D-R-D-R-D, R-D-R-D-R-D-R-D
4	default	0	Alternating D-R	$^(D-R)+\$$	2 locations per set	-
5	default	0	Alternating R-D-R	$^(R-D-R)+\$$	3 locations per set	-
6	default	0	Alternating D-R-D	$^(D-R-D)+\$$	3 locations per set	-
7	default	0	R^n ($n \geq 1, m = 0$)	$^R(?:-R)^*\$$	whole formation = 1 set	-

Appendix B: points to confirm with Aviral

Set id inside R^n-D^m . Four contracts in the mixed slice (F176 and F513 as R-R-D-D; F182 and F337 as R-R-R-D-D-D) are admitted by the R^n-D^m pattern, so all their rows share set 1. The alternative reading, pairing them positionally (first R with first D, and so on), was not discussed on the call. The National Grid demo slice is unaffected: every contract there is a plain R-D.

Alternating regexes in the dashboard. As written, $^(R-D)+\$$ only matches a dash-joined formation of exactly one R-D. Stage 4 now ignores dashes so the configured rows work as intended, but the cleaner fix is in the dashboard seed: $^R-D(?:-R-D)^*\$$ and likewise for the other three alternating rows.

Storage locations. WR / SB / IJ rows are excluded from stage 4 entirely. If every location of a contract should be visible in this table, they would be added as unpaired rows with no set id.

Unnamed segment ends in Transco-F2 and Transco-F3 (Angola LNG). Transco posted these six-location contracts as S9-S8-S9-S8-S9-S8, but only the first segment is real: Moss Point M3593 to Sta 85 Pool ZN4A carrying the full contract quantity (467,390 Dth). Positions 3, 4 and 5 have location code 0, no name and a zero or empty quantity, and position 6 is Sta 85 Pool ZN4 at zero quantity. Stage 4 pairs them positionally as sets 2 and 3 because that is what the alternating pattern says. Confirm whether such zero-quantity, unnamed pairs should stay PAIRED, be flagged, or be dropped.

A clean positional check: Transco-F1 (AGY Holding). Four receipt pools each pair with Carolina Gas as sets 1 to 4, and the four delivery quantities (58, 85, 65, 133) sum to the contract quantity of 341. This is the S8/S9 rule and the alternating set id working exactly as described on the call.